

Gooding Paul

Applied Research Scientist | AI & Engineering Innovator

Profile

Applied Research Scientist with expertise spanning applied AI, software engineering, hardware engineering, and business innovation. Experienced in applying cutting-edge research to solve real-world problems across healthcare, education, and community systems. Strong foundation in mathematics, embedded systems, and machine learning, with proven ability to lead multidisciplinary teams from concept to deployment.

Core Skills

- Programming: Python, C
- Applied Mathematics & Statistics
- Machine Learning & Deep Learning
- Hardware & Embedded Systems Design
- Research & Technical Writing
- Communication & Leadership

Professional Experience

Founder & Lead Researcher

That Blank Engineering Research Company — 2023 – Present

Founded and lead a multidisciplinary team of engineers and researchers to design and implement applied AI and hardware solutions for academic and community use

The startup was a passion project where the main goal was to have fun solving dynamic problems in different fields. This involves actively recruiting lectures, students and friends in order to solve prototyping challenges for our clients.

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- Designed and built a digital AI-powered microscope used to analyze and classify white blood cells.

The goal for this was to see if you can go about developing a low cost digital microscope that can analyse and count white blood cells from a sample.

I got to work with hematologists and phlebotomists in the project, in addition to that I got to work with mechanical designers and industrial designers in designing the overall microscope. I was involved in training and deploying the white blood cell counting ML onto the overall rig.

- Developed software to convert medical images into 3D-printable formats, enabling tangible visualization of radiology scans.

As part of my final year project, I ended up doing research on reconstructing radiological scans into 3D printable format. This project led me working hand in hand with a radiologist on ways to improve the overall quality of the prints where the main goal was to have said print be presented to visually impaired mothers and families.

- Engineered an early flood alert system, integrating sensors and communication protocols to provide real-time community alerts.

Got to work with RedCross and IRFC in developing an early warning system for Flood detection and mitigation. The project involved designing the overall architecture and ensuring that the project can work offgrid. The system is currently deployed at Tana River and can be accessed online.

- Developed the AI Pin, an assistive wearable device that leverages vision and audio feedback to communicate with the user.

Got to work in designing an AI powered digital assistance inspired by the Humane project. The project involves designing a product that incorporates only a microphone, speakers and camera. What ended up being produced was a handheld device you can communicate with and get response and feedback from it.

- Developed hospital software for Matt Xray Hospital that converts DOC/DOCX files into PDFs, digitally signs them, and automatically sends them via email.

-Developed PichaPrintAI for sketch to 3D design.

The aim behind PichaPrintAI is to see whether you can lower the barrier of entry in 3D design to the point that kids can take part in it. I was working in inco-operating doodle and sketches in the workflow such that kids simply upload a picture and a description of what they are making in their local dialect and have a 3D printable file at the end.

Biomed Intern

St. Francis Community Hospital — Jan 2025 – Sept 2025

- **Researched and developed Sheet Talk, an AI-powered voice assistant for Microsoft Excel that enables users to execute spreadsheet tasks via natural speech, lowering the learning curve and improving accessibility.**

While attached I observed that people still don't know how to use the excel in their inventory management , I ended up making an extension that allows them to simply chat with the excel with their local language and have it execute their command inside the excel document. This ended up cutting their cost of inventory management from 3 months to a single day.

- **Collaborated with hospital IT and administrative staff to test and refine the tool for healthcare data workflows.**

In addition to that i further refined their manual pen to paper to excel by developing a simple script that enables them to quickly upload to their inventory excel database through their phones and it automatically saves on time updating their inventory.

Educational Background

Technical University of Mombasa

B.Sc. Medical Engineering — 2018 – 2025

- Coursework: Biomedical Signal Processing, Embedded Systems, Machine Learning Applications in Healthcare, Clinical Instrumentation.
- Final Year Project: 3D Printing of Medical Images — developed a pipeline to convert radiology scans into 3D-printed models for improved visualization and accessibility in medical education and patient care.

Tenwek High School

Kenya Certificate of Secondary Education (KCSE) — 2014 – 2017

- Active in the Science and Engineering Club; developed early prototypes in electronics and coding projects, where I developed radio transmitters and receivers.
- Was part of Electrical and Electronics classes

Additional Highlights

- Strong research orientation with interest in AI for healthcare, assistive technologies, and biomedical instrumentation.
- Experience in leading cross-functional teams, project prototyping, and engaging in early business development.
- Passionate about bridging AI research with practical applications in communities and healthcare.